

COUNTEREXAMPLES TO THE EXACT-DISTANCE FORM OF CONJECTURE 6.13 IN SULLIVAN'S SURVEY

ABSTRACT. Sullivan's survey defines $N_j^+(v)$ as the set of vertices at directed out-distance exactly j from v and records Conjecture 6.13, attributed to an unpublished manuscript of Serge Burckel: a simple digraph with no directed cycle of length at most k has a vertex v satisfying $|N_k^+(v)| \geq |N_{k-1}^+(v)|$. We show that this printed exact-distance formulation is false for every $k \geq 3$. Indeed, $\text{Cay}(\mathbb{Z}_{3k-1}, \{1, 3\})$ has directed girth $k + 1$, while every vertex v satisfies $|N_{k-1}^+(v)| = 3 > |N_k^+(v)| = 2$. The same family satisfies the later recursively defined higher-order-neighborhood formulation.

For vertices u, v of a digraph D , let $d_D(u, v)$ be the length of a shortest directed path from u to v , and set

$$N_j^+(v) = \{u \in V(D) : d_D(v, u) = j\}.$$

Sullivan [1] explicitly uses this exact-distance convention and prints Conjecture 6.13 in the form stated in the abstract. A later paper of Daamouch [2] instead presents the Burckel-attributed question using recursively iterated out-neighborhoods. We have not seen Burckel's unpublished manuscript and make no claim about the formulation intended there. The result below concerns precisely the statement printed in Sullivan's survey.

Theorem 1. *For every integer $k \geq 3$, let $n = 3k - 1$ and*

$$D_k = \text{Cay}(\mathbb{Z}_n, \{1, 3\}),$$

with arcs $x \rightarrow x + 1$ and $x \rightarrow x + 3$ for each $x \in \mathbb{Z}_n$. Then D_k is a strongly connected, vertex-transitive oriented graph with indegree and outdegree 2 and directed girth $k + 1$. Moreover, every $v \in V(D_k)$ satisfies

$$|N_{k-1}^+(v)| = 3 \quad \text{and} \quad |N_k^+(v)| = 2.$$

Consequently, the exact-distance statement printed as Conjecture 6.13 in Sullivan's survey is false for every $k \geq 3$.

Proof. Translations of $\mathbb{Z}_n$ act transitively on D_k , and the generator 1 gives strong connectivity. Every vertex has indegree and outdegree 2. Since $n \geq 8$, the increments 1 and 3 are distinct and nonzero, and

$$\{1, 3\} \cap \{-1, -3\} = \emptyset \quad \text{in } \mathbb{Z}_n.$$

Thus the digraph is oriented.

2020 *Mathematics Subject Classification.* 05C20, 05C25, 05C38.

Key words and phrases. directed distance, directed girth, Cayley digraph, second neighborhood conjecture.

A directed walk of length ℓ using b arcs of increment 3 has unreduced displacement

$$\ell + 2b, \quad 0 \leq b \leq \ell.$$

If $1 \leq \ell \leq k - 1$, then

$$0 < \ell + 2b \leq 3\ell \leq 3k - 3 < n,$$

so the walk is not closed. If $\ell = k$, then

$$0 < k + 2b \leq 3k = n + 1 < 2n.$$

Closure would therefore require $k + 2b = n$, which is impossible because $k + 2b$ and $n = 3k - 1$ have opposite parity. Hence D_k has no directed cycle of length at most k . On the other hand, the walk whose increments are two copies of 1 followed by $k - 1$ copies of 3 is closed, since

$$2 + 3(k - 1) = n.$$

Its nonzero partial sums are 1, 2, and $2 + 3j$ for $1 \leq j \leq k - 2$; they are distinct integers strictly between 0 and n . Thus this walk is a directed cycle of length $k + 1$, and the directed girth is $k + 1$.

By translation, it remains to compute distances from 0. A walk of length d ends at

$$d + 2b \pmod{n}, \quad 0 \leq b \leq d.$$

Fix $2 \leq d \leq k - 1$. Since $3d < n$, none of these walks wraps modulo n . If $b \leq d - 3$, then

$$d + 2b = (d - 2) + 2(b + 1),$$

so the endpoint is already reachable in $d - 2$ steps. The remaining endpoints are

$$3d - 4, \quad 3d - 2, \quad 3d.$$

A walk of length at most $d - 2$ has displacement at most $3d - 6$, while every $(d - 1)$ -step displacement has parity $d - 1$ and the three displayed integers have parity d . Hence all three have distance exactly d , and

$$N_d^+(0) = \{3d - 4, 3d - 2, 3d\}, \quad 2 \leq d \leq k - 1.$$

In particular, $|N_{k-1}^+(0)| = 3$.

At length k , the endpoints with $b \leq k - 3$ were already reached in $k - 2$ steps. The remaining endpoints are

$$3k - 4 = n - 3, \quad 3k - 2 = n - 1, \quad 3k = n + 1 \equiv 1 \pmod{n}.$$

The residue 1 has distance 1. A walk of length at most $k - 2$ has unreduced displacement at most $n - 5$. A $(k - 1)$ -step walk has displacement at most $n - 2 < n$ and parity $k - 1$, whereas both $n - 3 = 3k - 4$ and $n - 1 = 3k - 2$ have parity k . Thus neither residue is reachable in fewer than k steps, and

$$N_k^+(0) = \{n - 3, n - 1\}.$$

Translation completes the proof. $\square$

Remark 2 (Recursive formulation). Daamouch [2] defines, for $A \subseteq V(D)$,

$$R^+(A) = \left(\bigcup_{x \in A} N^+(x) \right) \setminus A, \quad R_1^+(v) = N^+(v), \quad R_j^+(v) = R^+(R_{j-1}^+(v)).$$

The same graphs are not counterexamples under this convention. For $1 \leq j \leq k$, set

$$S_j = \{j + 2b \pmod{n} : 0 \leq b \leq j\}.$$

Since $2j < n$, the residues in S_j are distinct, so $|S_j| = j + 1$. We have $R_1^+(0) = S_1$, and for $2 \leq j \leq k$,

$$S_{j-1} + \{1, 3\} = S_j.$$

Moreover, $S_{j-1} \cap S_j = \emptyset$. Indeed, an element in the intersection would give

$$1 + 2(b - c) \equiv 0 \pmod{n}$$

for some $0 \leq b \leq j$ and $0 \leq c \leq j - 1$, but this integer is odd and

$$0 < |1 + 2(b - c)| \leq 2j + 1 \leq 2k + 1 < n.$$

Induction gives $R_j^+(0) = S_j$ for $1 \leq j \leq k$. Therefore, at every vertex v ,

$$|R_k^+(v)| = k + 1 > k = |R_{k-1}^+(v)|.$$

Thus Theorem 1 refutes only Sullivan's printed exact-distance formulation, not the later recursive formulation.

REFERENCES

- [1] B. D. Sullivan, *A summary of problems and results related to the Caccetta–Haggkvist conjecture*, AIM Preprint 2006-13, arXiv:math/0605646, 2006.
- [2] M. Daamouch, *A note on the second neighborhood problem*, Utilitas Mathematica **123** (2025), 155–164, doi:10.61091/um123-12.